

## Lesson 9 Assignment Due Monday Oct 12, 2026

### Instructions

Copy and paste SAS code into Canvas to submit your answers.

### Problem 1

We have collected the following data on five subjects:

| ID | AGE | GENDER | Grade Point Average (GPA) | College Entrance Exam Score (CSCORE) |
|----|-----|--------|---------------------------|--------------------------------------|
| 1  | 18  | M      | 3.7                       | 650                                  |
| 2  | 18  | F      | 2.0                       | 490                                  |
| 3  | 19  | F      | 3.3                       | 580                                  |
| 4  | 23  | M      | 2.8                       | 530                                  |
| 5  | 21  | M      | 3.5                       | 640                                  |

- (a) Write the SAS statements necessary to create a SAS dataset.
- (b) Add the statement(s) necessary to compute the mean grade point average and mean college entrance exam score.
- (c) We want to compute an index for each subject, as follows:

$$\text{INDEX} = \text{GPA} + 3 \times \text{CSCORE}/500$$

Modify your program to compute this INDEX for each student and to print a list of students in order of increasing INDEX. Include in your listing the student ID, GPA, CSCORE, and INDEX.

### Problem 2

Given the following set of data:

| Social Security Number | Annual Salary | Age | Race | Hire Date  |
|------------------------|---------------|-----|------|------------|
| 123874414              | 28,000        | 35  | W    | 4/12/1971  |
| 646239182              | 29,500        | 37  | B    | 3/15/1992  |
| 012437652              | 35,100        | 40  | W    | 4/15/1983  |
| 018451357              | 26,500        | 31  | W    | 12/10/2002 |

- (a) Create a SAS data set using the data above. Compute the average annual salary and average age. **NOTE:** Enter the annual salary values into DATALINES with commas.
- (b) If all subjects were in a 30% tax bracket, compute their taxes (based on annual salary) and print out a list, in Social Security number order, showing the annual salary, the tax, and the hire date.

### Problem 3

- (a) Import the dataset [https://manctsui.github.io/datasets/billboard\\_2021\\_01.csv](https://manctsui.github.io/datasets/billboard_2021_01.csv) and sort data by weeks on board.
- (b) To this dataset, write in a new column called `peak.position` which is a 1 if the rank equals peak rank and a 0 otherwise.